

Contribution Journal

Group Members Names:

Edwin Alvarado, Kala Hayes, Anashrah Adil, Zain Ahmed

Class: 6263-ITAI-1371 Intro to Machine Learning-S10-12321

Edwin

For this lab I completed the three data preparation tasks in the notebook. In Task 1, I handled the missing values by imputing the 177 missing 'Age' entries with the median, then verified that 0 missing values remained. In Task 2, I used one-hot encoding (`pd.get_dummies()`) to convert the 'Sex' and 'Embarked' columns into numeric columns and confirmed the output table. In Task 3, I applied `StandardScaler` to the 'Age' and 'Fare' columns to put them on the same scale. I ran and reviewed the output at each step to make sure the data preparation pipeline worked correctly in order. As well as writing the reflective journal.

Kala Hayes

I completed part 1 of the lab by setting up the coding environment and correcting the formatting of the code after the initial upload. The upload caused several lines of code to lose their proper spacing and indentation, resulting in syntax errors. I reformatted the code, ensured the dataset was imported correctly, and ran each cell to verify that it executed successfully. In Part 4, I implemented the feature scaling task by using `StandardScaler` to standardize the Age and Fare columns. After writing the code, I tested and troubleshooted it to ensure the numerical features were scaled correctly and that the code executed without errors.

Anashrah Adil

I ran all the cells after my team members completed their parts and double checked for any errors. I answered all the questions in the Knowledge check section to test my understanding of this lab. I researched about why median is a better choice than mean for imputation, details about One-Hot Encoding and how Decision Tree splits work.